

HW-01b — Conceptual Model Critique

Module M02 | Chapter 3 | Weight 3% | Difficulty ★

Module: M02 Chapter: 3 Weight: 3% Due: See course schedule

Prerequisite: You must have submitted HW-01 (your own conceptual model) before completing this assignment.

Overview

A conceptual model is only as good as its reviewability. Professional simulation analysts peer-review each other's models before writing any code — errors caught at the conceptual stage cost nothing; errors found after a model is built cost weeks.

In this assignment you will critique **one** provided model, identify its gaps, and then revise your own HW-01 model in response to the same checklist.

1 Provided Model: Airport Security Checkpoint

Read the following conceptual model document and answer the critique questions in Part A.

Draft Conceptual Model

Narrative. An airport security checkpoint processes passengers before they enter the secure terminal. Passengers arrive, pass through an ID check, walk through a body scanner, collect their belongings at a reclaim belt, and proceed to the gate. The simulation will answer: “How many screening lanes minimise mean passenger throughput time while keeping queue lengths below 20?”

Entities.

Entity	Attributes
Passenger	Arrival time

Resources.

Resource	Capacity	Service time	Discipline
ID checker	1 per lane	Fast	FCFS
Body scanner	1 per lane	Variable	FCFS
Reclaim belt	1 per lane	Varies	FCFS

Events.

- Passenger arrives
- Passenger departs

State variables.

- Number of passengers in the checkpoint

Performance measures.

- Average throughput time
- Queue length

Assumptions.

1. Arrivals are random.
2. All passengers eventually pass through.
3. The checkpoint is busy during the day.

2 Part A: Critique the Provided Model (50 pts)

Answer each question with reference to specific deficiencies in the draft above. For each issue you identify, quote the problematic section and explain what is missing or incorrect.

2.1 A1 — Entity completeness (8 pts)

The entity table lists only one attribute (“Arrival time”) for passengers.

- (a) List at least **four** additional attributes that a real checkpoint model would need to distinguish passenger behaviour or routing.
- (b) Are there other entity types missing entirely? Identify at least one and explain what role it plays.

2.2 A2 — Resource specification (10 pts)

The resource table is underspecified.

- (a) Replace “Fast,” “Variable,” and “Varies” with plausible probability distributions (name the distribution and state typical parameter values in minutes based on general knowledge of airport security).
- (b) The capacity column says “1 per lane” but the performance question asks about “number of screening lanes.” Explain how the number of lanes relates to resource capacity in a DES model. Rewrite the resource table to correctly represent a 3-lane checkpoint.

2.3 A3 — Event completeness (10 pts)

The event list contains only two events. A real checkpoint has at least eight distinct events. List all events, state what state change each causes, and identify which resources (if any) are seized or released at each event.

2.4 A4 — State variable completeness (8 pts)

“Number of passengers in the checkpoint” is a single aggregate variable. List at least **five** state variables needed to faithfully represent the checkpoint state — enough to resume the simulation from any snapshot.

2.5 A5 — Assumptions (8 pts)

The three assumptions are vague and incomplete.

- (a) For each provided assumption, rewrite it as a precise, quantified statement (e.g., “Arrivals are random” → “Passenger interarrival times follow an Exponential distribution with mean 12 seconds”).
- (b) Add at least **four** new assumptions not present in the draft. For each new assumption, state what modelling error it introduces if the assumption is violated.

2.6 A6 — Scope and performance measures (6 pts)

The performance question asks about minimising throughput time while keeping queue lengths below 20.

- (a) Is this a well-posed question for DES? Identify what is ambiguous.
- (b) Rewrite the question as a precise simulation study objective, naming the decision variable, the objective function, and the constraint.

3 Part B: Revise Your Own HW-01 Model (35 pts)

Apply the same critique checklist to your own HW-01 conceptual model. Produce a **revised version** that addresses any deficiencies.

3.1 B1 — Revised model document (10 pts)

Submit the revised model document. It must now meet all eight components from Section 3.5 of the textbook: narrative, flow diagram, entity table, resource table, event list, state variable list, performance measure list, and assumptions.

3.2 B2 — Change log (15 pts)

Write a change log (bullet list) explaining every revision you made relative to your HW-01 submission. For each change, state:

- (a) what was missing or wrong,
- (b) what you changed, and

(c) why the revision makes the model more credible.

Minimum: five substantive changes.

3.3 B3 — Flow diagram (10 pts)

Your revised model must now include a flow diagram. This can be hand-drawn and photographed, or drawn with any digital tool (draw.io, PowerPoint, etc.).

Requirements:

- Entities are represented by tokens or shapes.
- Resources are labelled.
- Decision points (routing) are marked with probabilities or conditions.
- Entry and exit points are labelled.

4 Part C: Reflection (15 pts)

Answer in 2–4 sentences each.

4.1 C1 (5 pts)

What is the most consequential modelling decision in a conceptual model — the one whose error would most damage the validity of any simulation built from it? Defend your choice.

4.2 C2 (5 pts)

The airport checkpoint model omits passengers with accessibility needs, who may require additional screening time or a separate lane. Is this a simplifying assumption or a boundary decision? What would you need to know (data or domain knowledge) before deciding whether to include it?

4.3 C3 (5 pts)

Can a conceptual model ever be “complete”? What stops a modeller from adding more and more detail indefinitely, and what principle should govern when to stop?

Deliverables

Combine all parts into a single PDF (≤ 6 pages excluding the flow diagram). Submit via GitHub Classroom.

Item	Format
Part A: critique (A1–A6)	In-document, clearly labelled
Part B: revised model document	Updated document
Part B: flow diagram	Embedded or appended
Part B: change log	Clearly labelled bullet list
Part C: reflection (C1–C3)	Short answers

Grading Summary

Component	Points
A — Critique (A1–A6)	50
B — Revised model	35
C — Reflection	15
Total	100

Full rubric: [rubrics/homework_rubric.pdf](#) | Exemplar critiques: the instructor package (available to instructors on request).